Consistency Center-Based Deep Cross-Modal Hashing for Multisource Remote Sensing Image Retrieval

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***Abstract*— Cross-modal hashing aims to retrieve similar images from large-scale Earth observation (EO) data archives, which typically contain multiple satellite sources of remote sensing (RS) images. However, existing cross-modal hashing methods primarily focus on dual-source RS images and often face two main limitations when retrieving multisource RS images. First, these methods exhibit significant redundancy as they require handling all possible dual-source combinations in multisource RS images. Second, they often rely on pairwise or triplet image sources to construct objective functions, which are not significantly effective in reducing the discrepancies among multiple RS image sources. To address these limitations, we propose a novel consistency center-based deep cross-modal hashing method for multisource RS image retrieval called C^{2}Hash. Our C^{2}Hash employs a multibranch hashing network to directly encode multisource RS images into unified hash codes, thereby offering higher processing efficiency. Furthermore, C^{2}Hash introduces consistency centers to construct a novel objective function. The consistency center represents the shared semantic features among similar multisource RS images and is generated by a label hashing network. The objective function encourages similar multisource RS images to approach the same consistency center to align all image sources in a unified Hamming space. Our method can effectively reduce the discrepancies across multiple image sources and generate unified hash codes. To evaluate its effectiveness, we construct a new multisource RS image dataset called MSRSI, comprising five different types of image sources. We conduct comprehensive experiments to demonstrate the superior performance of our method on the MSRSI dataset (https://github.com/sunyuxi/C2Hash).**

***Index Terms*— Content-based remote sensing (RS) image retrieval, deep supervised hashing, learning to hash, multisource remote sensing images.**

I. Introduction

With the development of Earth observation (EO) technologies, we now have access to vast archives of EO data [1], [2]. These archives comprise extensive and diverse remote sensing (RS) images, such as very high-resolution (VHR) optical images, panchromatic (PAN) images, and synthetic aperture radar (SAR) images. Retrieving images of interest from these EO data archives is of significant importance for a variety of applications, such as disaster rescue [3] and economic assessment [4]. Consequently, several RS image retrieval tasks have been proposed, including image-to-image retrieval [5], [6], [7], [8], [9], [10], [11], [12], audio-to-image retrieval [13], [14], [15], [16], [17], and text-to-image retrieval [18], [19], [20], [21], [22]. This article focuses on exploring the image-to-image retrieval task, which involves finding similar images to a query image from an RS image database. As stated in [1], learning to hash is an efficient solution for large-scale image-to-image retrieval. This approach encodes RS images into compact hash codes by learning nonlinear hashing functions. Based on hash codes, image-to-image retrieval can quickly and accurately find similar images from vast EO data archives.

Generally, existing learning-to-hash algorithms can be categorized into unimodal hashing methods [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36] and cross-modal hashing methods [7], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47]. Unimodal methods utilize RS images from a single satellite source to learn hashing functions. For example, Li et al. [23] developed a deep hashing network to generate discriminative hash codes for optical RS images. Chen et al. [25] proposed a deep saliency smoothing hashing method for drone images, which incorporates a saliency capture mechanism, a distribution smoothing term, and global and local features to learn effective hash codes.

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In contrast, cross-modal hashing methods employ different satellite sources of RS images to generate nonlinear hashing functions. These images often exhibit significant domain differences in modalities. The goal of cross-modal hashing is to encode different modality data into a unified Hamming space to reduce intermodal discrepancies [37], [38]. For example, Li et al. [37] utilized a pairwise similarity constraint loss function to design a source-invariant deep hashing network for PAN and multispectral image (MSI) retrieval. Xiong et al. [38] proposed a cross-modal hashing network with a novel triplet loss to minimize modality differences between MSI and SAR images. These cross-modal hashing methods primarily focus on dual-source RS images and have demonstrated promising performance in dual-source RS image retrieval tasks.

However, EO data archives typically contain multiple types of satellite sources [48], [49], such as VHR, SAR, PAN, and MSI. Researchers rarely explore deep cross-modal hashing for multisource RS images. Moreover, existing cross-modal hashing usually cannot be effectively applied to multisource RS image retrieval due to the following three reasons.

1) There are a few datasets available for multisource RS image retrieval. Existing datasets mainly focus on dual-source RS image retrieval. For example, the DSRSID [37] dataset contains both PAN and MSI images.
2) Current hashing methods only handle dual-source images, whereas multisource RS images encompass numerous combinations of dual-source images. As a result, current hashing methods become complex and redundant when employed for multisource RS image retrieval. This is because these methods require numerous additional hashing models and memory to handle all possible dual-source combinations. For example, a set of five-source images $[S_1, \ldots, S_5]$ can produce ten ($C_5^2 = 10$) unique combinations of dual-source images (e.g., S_1 and S_2), which would require 20 ($A_5^2 = 20$) hashing models and 20 hash tables to match images between any sources.
3) Multisource RS images exhibit a greater number of modal discrepancies. Deep hashing needs to employ objective functions to address these discrepancies. Existing objective functions mainly utilize pairwise or triplet constraints to deal with two modalities. When confronted with multisource RS images, these functions are unable to effectively align all modalities within a unified Hamming space. This is because they only handle the dual-source discrepancies in multisource RS images. As a result, existing objective functions cannot effectively deal with multiple modal discrepancies when encoding multisource RS images into unified hash codes.

To address these challenges, we first construct a novel multisource RS image dataset called MSRSI. This dataset comprises 150 000 images collected from 15 land-cover categories and five types of RS image sources: VHR, SAR, MSI, mean average precision (MAP), and PAN. Fig. 1(a) provides an example of a river image from our dataset. Second, we propose a novel consistency center-based deep cross-modal hashing method for multisource RS image retrieval, named C^{2}Hash. Our method comprises a deep multibranch hashing

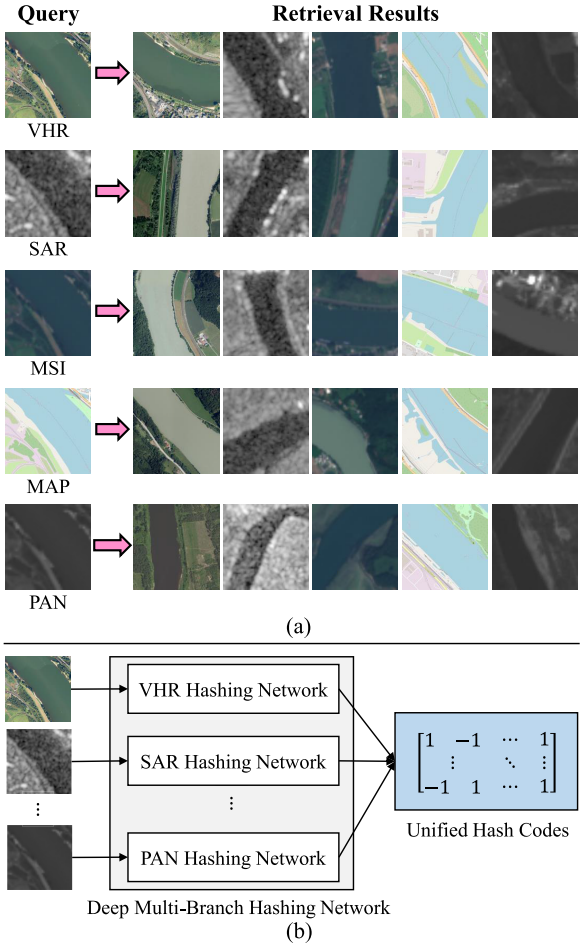

Fig. 1. Example of (a) multisource RS image retrieval and (b) deep cross-modal hashing for multisource RS images.

network, as shown in Fig. 1(b), along with a label hashing network. The multibranch network directly encodes all multisource RS images into unified hash codes during the training phase. In the test phase, each branch network independently extracts hash codes for each image source. Our method only requires one hash model and one hash table for each image source, which is more efficient and flexible compared to existing hashing techniques. These techniques typically require A_n^2 hash models and A_n^2 hash tables for n image sources, which means $n-1$ hash models and $n-1$ hash tables per image source. Third, we design a novel objective function based on consistency centers in our proposed method. These consistency centers are generated by the label hashing network using class labels. Each center represents the common semantic features shared by similar multisource RS images in the Hamming space. The objective function encourages similar multisource RS images to approach the same consistency center to reduce the modal discrepancies across multiple image sources.

Specifically, the objective function consists of a center-based similarity preserving loss, a multisource hashing consistency loss, and a hash quantization loss. The similarity preserving loss combines consistency centers to learn semantic similarity within the Hamming space, thereby preserving semantic coherence across different modalities. The hashing consistency loss minimizes the distance between multisource RS images

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and their corresponding consistency centers to reduce modal discrepancies. The hash quantization loss balances the distribution of hash codes. Our method utilizes the objective function to align all modalities in a common Hamming space and generate unified hash codes for multisource RS images. We have evaluated our newly proposed method on the MSRSI dataset and demonstrated its effectiveness for multisource image retrieval.

The contributions of this article are summarized as follows.

1) We introduce a novel multisource RS image retrieval problem that focuses on five different RS image sources. To the best of our knowledge, this problem has not been previously explored, presenting a new challenge to the RS community.
2) We propose a consistency center-based deep hashing method, called C^2Hash, which encodes multisource RS images into unified hash codes. In the proposed method, we design a novel consistency center-based objective function, which consists of a center-based similarity preserving loss, a multisource hashing consistency loss, and a hash quantization loss. The objective function utilizes consistency centers to align all RS image sources and reduce their modal discrepancies in a common Hamming space.
3) We have constructed a large-scale multisource RS image dataset, named MSRSI, comprising VHR, MSI, SAR, MAP, and PAN images. We have also made this dataset publicly available to facilitate the study of multisource RS image retrieval. Furthermore, we have conducted comprehensive experiments on this dataset to demonstrate the effectiveness of our method.

The rest of this article is organized as follows. Section II provides a brief review of the related work. Section III presents a detailed description of our method. In Section IV, we introduce our constructed dataset and the experimental results in detail. Section V concludes this article with a summary.

II. Related Work

In this section, we conduct a literature review of unimodal hashing methods and cross-modal hashing methods.

A. Unimodal Hashing Methods

Unimodal hashing methods [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36] learn discriminative binarized features in Hamming space by preserving the similarity between RS images captured by the same satellite sensor. For example, Li et al. [23] proposed a deep hashing network to capture the rich semantic information of RS images and learn hash functions by utilizing the semantic similarity between image pairs. To address storage and computational constraints, Li et al. [24] introduced a quantized deep hashing framework where both the weights and activation functions are binarized to low-bit representations. To retrieve similar drone images, Chen et al. [25] presented a deep saliency smoothing hashing method that effectively learns hash codes by incorporating a saliency capture mechanism, distribution smoothing term, and global and local features. These methods utilize symmetric hash functions for both query and database images, which may be time-consuming when dealing with large-scale databases. In contrast, Song et al. [26] proposed an asymmetric hashing approach that generates hash codes for query images by binarizing the output of hashing networks, while the hash codes for database images are directly learned by solving the objective function.

However, the performance of deep unimodal hashing methods can be limited when the number of labeled images is insufficient. To address this limitation, Tang et al. [27] introduced a meta-learning-based supervised hashing method that leverages a small number of labeled training samples to train hashing networks. Hou et al. [28] utilized labeled data from a source domain to learn hash functions for unlabeled data in a target domain through unsupervised domain adaptation algorithms. Nonetheless, these methods are limited to unisource RS images and cannot handle cross-source RS images. In practical applications such as all-weather rescue [50], the use of cross-source RS images is necessary, and unimodal methods cannot be directly applied to learn cross-source hashing functions [38]. Consequently, there is a growing interest in cross-modal hashing methods.

B. Cross-Modal Hashing Methods

Cross-modal hashing [7], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [51], [52], also known as cross-source hashing, utilizes cross-modal RS data to learn nonlinear hashing functions. Cross-source RS images usually exhibit significant visual differences [53], [54]. Cross-modal hashing needs to reduce intermodal discrepancies and encode cross-modal data into hash codes in a common Hamming space. For example, Li et al. [37] developed a source-invariant deep hashing network with a series of optimization constraints to retrieve PAN and MSI images. Xiong et al. [38] designed a cross-modal hashing network and employed a new training strategy to eliminate the cross-modal discrepancies for optical and SAR image retrieval. In addition, Xiong et al. [5] proposed an image translation-based framework that utilizes a cycle-identity-generative adversarial network to map images from the source domain to the target domain for cross-source RS image retrieval. Fang et al. [7] utilized interclass pairwise disperse cues and intraclass pairwise balance cues to construct separate hash code books for sketches and RS images. Sun et al. [46] introduced a deep hashing method based on multisensor fusion and explicit semantic preservation for SAR and VHR image retrieval. Guo et al. [51] developed a distillation-based hashing technique to effectively address the intricate geometric attributes present in vessel images. Ding et al. [52] introduced an end-to-end similarity metric learning network to effectively tackle the issue of data drift among cross-source RS images. However, these methods may face limitations when the number of labeled examples is insufficient. Chaudhuri et al. [47] presented a label-deficit zero-shot learning strategy and a bilevel Siamese network for cross-sensor data retrieval, effectively utilizing a small number of labeled data samples from known classes during training. The aforementioned hashing methods primarily focus on image-to-image retrieval.

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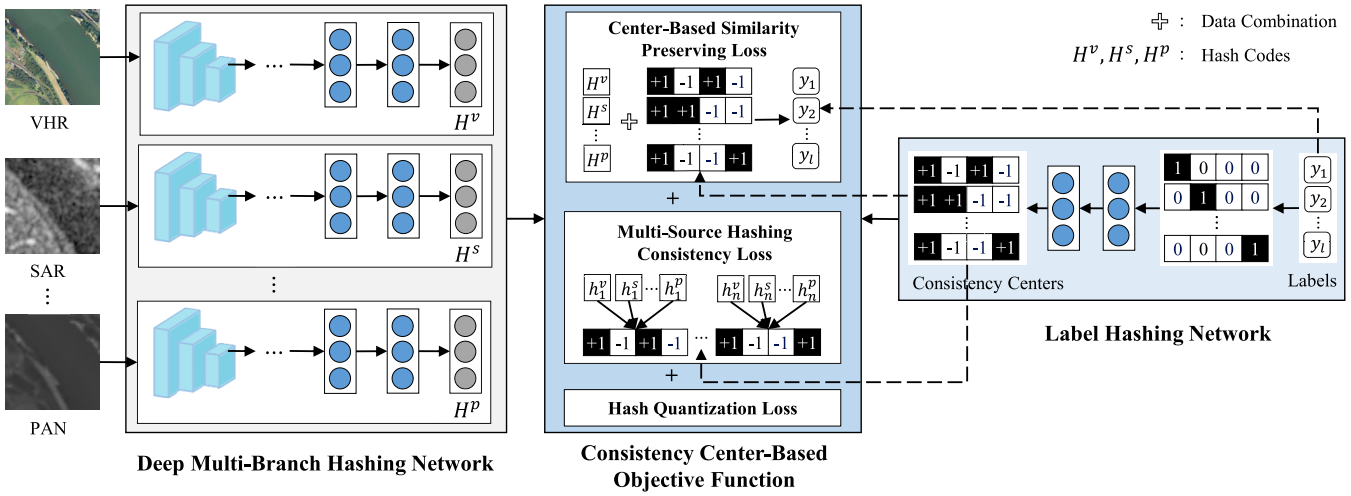

Fig. 2. Overview of the proposed C^{2}Hash method based on consistency centers for multisource RS image retrieval. C^{2}Hash consists of a deep multibranch hashing network, a label hashing network, and a consistency center-based objective function. H^v, H^s, and H^p denote the hash codes generated by the hash layers in the multibranch hashing network for VHR, SAR, and PAN images, respectively.

Cross-modal hashing techniques are widely employed in text-to-image and audio-to-image retrieval tasks. For example, Jiang and Li [43] integrated feature learning and hash code learning into an end-to-end deep learning framework for text and image retrieval. Jin et al. [45] devised a cross-modal hashing model that explores the correlation structure and semantic information across different modalities. Yuan et al. [18] developed an asymmetric multimodal feature-matching network for text-to-image matching. This network incorporates a multiscale visual self-attention module to extract salient features from RS images. These features are then used to guide text representation for cross-modal RS text–image retrieval. Yuan et al. [19] proposed a multilevel information dynamic fusion module that effectively integrates global and local information for text–image retrieval. Chen et al. [13] introduced a deep quadruple-based hashing approach that employs a quadruple construction hard module to learn relative semantic similarity relationships between audio and RS images. In addition, Chen et al. [17] proposed a fine-aligned discriminative hashing approach to capture discriminative information from RS images and learn the corresponding detailed information between RS images and audio.

Existing cross-modal hashing methods primarily focus on dual-source RS image retrieval. However, EO data archives typically consist of multiple types of RS image sources. Existing deep cross-modal hashing techniques are rarely explored for multisource RS image retrieval. In this article, we propose a consistency center-based deep cross-modal hashing method, which encodes multisource RS images into unified hash codes for efficient multisource RS image retrieval.

III. Method

In multisource RS image retrieval, we are provided with a dataset consisting of n instances $\mathcal{X} = \{(x_i^v, x_i^s, x_i^u, x_i^a, x_i^p, y_i)\}_{i=1}^n$, where x_i^v, x_i^s, x_i^u, x_i^a, and x_i^p represent the ith VHR optical image, SAR image, multispectral image, map image, and PAN image, respectively. The class label of the ith instance is denoted by $y_i \in \{1, 2, \ldots, l\}$, where l denotes the total number of class labels. For convenience in representing formulas, we utilize numerical superscripts to indicate these data sources sequentially. For example, x^1 corresponds to VHR images, while x^5 represents PAN images. The total number of image sources is denoted by M. The objective of deep cross-modal hashing for multisource RS image retrieval is to encode images from various data sources into K-bit hash codes within a common Hamming space by learning nonlinear hashing functions. These hash codes can be used to measure the similarity between images from different sources.

A. Overview

As shown in Fig. 2, we propose a consistency center-based deep cross-modal hashing for multisource RS image retrieval called C^{2}Hash, which comprises three main components: a deep multibranch hashing network, a label hashing network, and a consistency center-based objective function.

The deep multibranch hashing network consists of multiple independent hashing networks, each applied to an individual image source. These hashing networks do not share weights and are comprised of convolutional layers, fully connected layers, and a hash layer, which have a similar architecture to that of [37] and [46]. The purpose of each hashing network is to generate hash codes for its respective image source. The label hashing network is responsible for generating consistency centers, where each center denotes the shared semantic features among similar multisource RS images within the Hamming space. With these centers and labels, we design a consistency center-based objective function, which includes three loss components: a center-based similarity preserving loss, a multisource hashing consistency loss, and a hash quantization loss. In the proposed method, the label hashing network is exclusively employed during the training phase, while the multibranch hashing network is utilized for generating hash codes for input images in both the training and testing phases. In addition, each branch network has the

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capability to independently extract hash codes for their respective images.

B. Deep Multibranch Hashing Network

Deep cross-modal hashing for multisource RS image retrieval aims to generate hash codes to measure the similarity between different image sources. Existing hashing methods [37], [38], [46], [47] typically employ a dual-branch network to learn hash functions for different RS images. However, when dealing with multisource RS images, numerous combinations of dual-source images arise. Two-branch networks become inefficient and redundant as they require training additional hashing networks to learn all combinations and consume more memory to maintain more hash tables. Therefore, we propose a deep multibranch network to deal with multisource RS images. The multibranch network comprises multiple hashing networks and each hashing network corresponds to an image source. Moreover, these hashing networks do not share their weights with each other. During the training phase, all hashing networks are simultaneously trained to encode these multisource images into a common Hamming space. In the testing phase, each hashing network is independently used to extract the hash codes for each image source.

The objective of each hashing network is to efficiently encode the images from an image source into hash codes. Each hashing network contains a feature extraction network and a hash layer. As shown in Fig. 2, the high-dimensional feature of each image from different satellite sources is first extracted by a feature extraction network, which is defined as follows:

$$z_i^m = f_m\left(x_i^m, W_z^m\right) \tag{1}$$

where x_i^m represents the ith image from the mth image source, z_i^m denotes the high-dimensional feature of x_i^m, and f_m refers to the feature extraction network with the parameter W_z^m for the mth image source. Afterward, z_i^m is transformed into a compact hash code using a hash layer, as formulated by

$$h_i^m = \operatorname{sign}\left(g_m\left(z_i^m, W_h^m\right)\right), \quad h_i^m \in \{0, 1\}^K \tag{2}$$

with

$$\operatorname{sign}(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases} \tag{3}$$

where the full connection layer g_m with parameter W_h^m is responsible for converting z_i^m into a hash-like feature. The sign$(\cdot)$ function is an elementwise sign function that returns 1 if the element is positive and 0 otherwise. The resulting h_i^m represents the desired hash code of x_i^m.

However, the sign$(\cdot)$ function is nondifferentiable at zero, which prevents the hashing network from being updated using the backpropagation algorithm. In practice, we replace sign$(\cdot)$ with a tanh$(\cdot)$ function. Equation (2) is reformulated as follows:

$$h_i^m = \tanh(g_m(z_i^m, W_h^m)), \quad h_i^m \in \{-1, +1\}^K. \tag{4}$$

The tanh$(\cdot)$ function encourages each hash code to approach $+1$ and -1 during training. These values can then be transformed into 1 and 0 using a linear function, i.e., $y = (x+1)/2$.

C. Label Hashing Network

Existing cross-modal hashing methods typically rely on pairwise or triplet data modalities to learn binarized representations for dual-source images. However, when dealing with multisource RS images, these methods only capture local relationships between two source images. This limitation makes it challenging to learn discriminative hash functions for multisource RS images in a common Hamming space. To overcome this limitation, we propose to utilize consistency centers to learn unified hash codes for multisource RS images. These consistency centers denote shared binarized representations for each class across all multisource RS images, and they are positioned at a large Hamming distance from each other. Our approach encourages multisource RS images belonging to the same class to converge toward their respective center, while multisource RS images from different classes are scattered around different centers. By leveraging these centers, we can simultaneously learn the hash functions of the images from multiple satellite sources, rather than just a subset of dual-source images. This enables us to generate discriminative unified hash codes for multisource RS images within a common Hamming space.

We develop a label hashing network to generate consistency centers, which are subsequently incorporated into the objective function described below. This allows us to effectively capture semantic similarity and minimize discrepancies across multiple modalities. As shown in Fig. 2, the consistency centers are generated using the following equation:

$$C = f_c(L, W_c), \quad L \in \mathbb{R}^{l \times l}, \ C \in \mathbb{R}^{l \times K} \tag{5}$$

where L represents a label matrix containing class labels, C denotes learned consistency centers, and f_c refers to the label hashing network parameterized by W_c. In our implementation, we utilize a multilayer perceptron as the label hashing network. To ensure sufficient discrimination between consistency centers, we introduce an orthogonality constraint to the center matrix. In addition, a linear relationship between the Hamming distance $\operatorname{dist}_H(\cdot, \cdot)$ and the inner product $\langle \cdot, \cdot \rangle$ has been observed, as reported in [55]. The relationship can be defined as follows:

$$\operatorname{dist}_H(h_i, h_j) = \frac{1}{2}\left(K - \left\langle h_i, h_j \right\rangle\right) \tag{6}$$

where K represents the hash code length and h_i or $h_j \in \{-1, 1\}^K$ denotes a hash code.

Based on the relationship between Hamming distance and inner product, we define the orthogonality constraint for consistency centers as follows:

$$C^* = \min \left\| \frac{1}{K} C^T C - \mathbf{I} \right\|_2 \tag{7}$$

where $\mathbf{I}$ represents an identity matrix and C^* denotes the optimal consistency centers. Each consistency center corresponds to a class across multisource RS images. We encourage similar multisource RS images to approach the same center, while dissimilar RS images are encouraged to approach different centers. By utilizing these centers, we can preserve semantic similarity, eliminate discrepancies across multiple modalities,

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and generate discriminative unified hash codes for multisource RS images.

D. Consistency Center-Based Objective Function

C^{2}Hash aims to learn binarized representations of multisource RS images in a common Hamming space. This requires preserving semantic similarity while minimizing discrepancies across multiple modalities. We design an objective function based on consistency centers to optimize the entire network. The consistency centers have been generated by the label hashing network, with each center representing the shared semantic features among similar multisource RS images in the Hamming space. The objective function utilizes both consistency centers and class labels to generate unified hash codes for multiple image sources. Specifically, it ensures that similar multisource RS images have similar hash codes with a high probability, while dissimilar images have dissimilar hash codes. To accurately estimate this probability, we employ a Bayesian learning framework to derive the objective function. By leveraging class labels Y and generated consistency centers C, we define the logarithm maximum a posteriori estimation of hash codes for multiple image sources as follows:

$$\begin{aligned}\log \prod_{m=1}^{M} p(H^m|Y,C) &\propto \log \prod_{m=1}^{M} p(Y,C|H^m)p(H^m)\\ &= \log \prod_{m=1}^{M} p(Y|C,H^m)p(C|H^m)p(H^m)\\ &= \sum_{m=1}^{M}\sum_{i=1}^{n} \log p(y_i|c_i,h_i^m)p(c_i|h_i^m)p(h_i^m)\\ &= \sum_{m=1}^{M}\sum_{i=1}^{n} \log p(y_i|c_i,h_i^m)\\ &\quad + \sum_{m=1}^{M}\sum_{i=1}^{n} \log p(c_i|h_i^m)\\ &\quad + \sum_{m=1}^{M}\sum_{i=1}^{n} \log p(h_i^m) \end{aligned} \tag{8}$$

where H^m represents the hash codes of the images from the mth source. $p(y_i|c_i,h_i^m)$ denotes a likelihood function that utilizes consistency centers to preserve semantic similarity. $p(c_i|h_i^m)$ represents a hashing consistency likelihood function that reduces modal discrepancies by encouraging similar images across different modalities to converge toward the same centers. $p(h_i^m)$ represents the prior distribution for the learned hash codes of the mth image source.

$p(y_i|c_i,h_i^m)$ represents the conditional probability of the class label y_i given the hash code h_i^m and the consistency center c_i. By utilizing (5) and (7), we can infer that the center c_i tends to be distributed in the Hamming space and conveys the semantic information associated with the class label y_i. The center establishes a connection between the Hamming space and the semantic similarity. Therefore, we incorporate the center to preserve the semantic similarity of the hash codes belonging to class y_i across multiple modalities. To accomplish this, we employ an exponential distribution and integrate the orthogonal constraints of the consistency centers to formulate the probability $p(y_i|c_i,h_i^m)$ as follows:

$$\begin{aligned}\log p\left(y_i\middle|c_i,h_i^m\right) &= \log \gamma e^{-\gamma\left(\sigma\left(y_i,c_ih_i^m\right)+\left\|\frac{1}{K}C^TC-\mathbf{I}\right\|_2\right)}\\ &\propto -\left(\sigma\left(y_i,c_ih_i^m\right)+\left\|\frac{1}{K}C^TC-\mathbf{I}\right\|_2\right)\\ &= -\left(y_i\log\left(c_ih_i^m\right)+(1-y_i)\log\left(1-c_ih_i^m\right)\right.\\ &\quad \left.+\left\|\frac{1}{K}C^TC-\mathbf{I}\right\|_2\right)\end{aligned} \tag{9}$$

where σ is the sigmoid function. Through $p(y_i|c_i,h_i^m)$, the semantic information of similar images from different modalities can be preserved by hash codes.

$p(c_i|h_i^m)$ represents the conditional probability of c_i given h_i^m. Multisource RS images encompass various combinations of dual-source images, making it challenging to learn a unified binarized representation by directly aligning all these combinations. Instead of handling every dual-source combination, we encourage similar multisource RS images to approach the same consistency center to align different image sources. These centers are pretransformed into the Hamming space. When similar (or dissimilar) images from different modalities are close to the same (or different) center, it indicates that the differences across modalities are reduced. Therefore, we employ these centers to maintain hashing consistency for multisource RS images and utilize the exponential distribution to model $p(c_i|h_i^m)$. Furthermore, since c_i is a binarized vector with the same dimension as h_i^m, we transform the optimization problem of the hashing vector into a multihash bit optimization problem. The definition of $\log p(c_i|h_i^m)$ is given as follows:

$$\begin{aligned}\log p\left(c_i|h_i^m\right) &= \log \alpha e^{-\alpha\sigma\left(c_i,h_i^m\right)}\\ &= \log\alpha\prod_{j=1}^{K} e^{-\alpha\sigma\left(c_{ij},h_{ij}^m\right)}\\ &\propto -\sum_{j=1}^{K} c_{ij}\log h_{ij}^m+(1-c_{ij})\log\left(1-h_{ij}^m\right)\end{aligned} \tag{10}$$

where K represents the length of the hash code and σ denotes the sigmoid function. By incorporating consistency centers, $p(c_i|h_i^m)$ can effectively reduce the discrepancies across multiple modalities and generate unified hash codes within a common Hamming space.

$p(h_i^m)$ represents the prior probability of hash codes, where each bit of the hash codes is distributed around the values $\{-1, 1\}$. To encourage the hash bits to concentrate around $\{-1, 1\}$, we utilize a bimodal Laplacian prior distribution to model $p(h_i^m)$. The definition of $p(h_i^m)$ is given as follows:

$$\begin{aligned}\log p(h_i^m) &= \log\frac{1}{2\beta}e^{-\frac{\|\,|h_i^m|-\mathbf{1}\|_1}{\beta}}\\ &\propto -\|\,|h_i^m|-\mathbf{1}\|_1\end{aligned} \tag{11}$$

where $\mathbf{1}$ is a constant vector consisting of ones, $|\cdot|$ calculates the elementwise absolute value, and $\|\cdot\|_1$ denotes the $L1$ normalization. The bimodal Laplacian distribution assigns

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the highest density to the discrete values $\{-1, 1\}$, which encourages each hash bit to approach $\{-1, 1\}$ with the highest probability. However, (11) is nonsmooth. Therefore, we employ a smooth approximation of the absolute function [55]: $|x| \approx \log\cosh x$, which simplifies (11) as follows:

$$\begin{aligned}\log p(h_i^m) &\propto -\| |h_i^m| - \mathbf{1}\|_1 \\ &= -\log\cosh(|h_i^m| - \mathbf{1}).\end{aligned} \tag{12}$$

$p(h_i^m)$ can optimize the distribution of each bit to maximize the representation capacity of hash codes.

We take (9), (10), and (12) into the maximum a posteriori estimation in (8). Consequently, we derive a consistency center-based objective function defined as follows:

$$\min_{\Theta} \mathcal{L} = L_{\text{cs}} + \lambda_0 Q_{\text{hc}} + \lambda_1 Q_{\text{hq}} \tag{13}$$

where L_{cs} represents a center-based similarity preserving loss, Q_{hc} denotes a multisource hashing consistency loss, and Q_{hq} represents a hash quantization loss. λ_0 and λ_1 are hyperparameters that adjust the weighting assigned to Q_{hc} and Q_{hq}, respectively. The objective function $\mathcal{L}$ is minimized to optimize C^{2}Hash with respect to the parameter space Θ.

The center-based similarity preserving loss is derived from (9), which utilizes consistency centers to maintain semantic similarity across various modalities. The definition of the center-based similarity preserving loss is given as follows:

$$\begin{aligned}L_{\text{cs}} = \sum_{m=1}^{M}\sum_{i=1}^{n} y_i \log\left(c_i h_i^m\right) + (1 - y_i)\log\left(1 - c_i h_i^m\right) \\ + \left\| \frac{1}{K} C^T C - \mathbf{I} \right\|_2.\end{aligned} \tag{14}$$

The multisource hashing consistency loss is derived from (10), which encourages similar images from different modalities to converge toward the same centers to minimize modal discrepancies. When similar (or dissimilar) images from various modalities are located close to the same (or different) center, the differences across modalities are correspondingly diminished. The multisource hashing consistency loss is defined as follows:

$$Q_{\text{hc}} = \sum_{m=1}^{M}\sum_{i=1}^{n}\sum_{j=1}^{K} c_{ij}\log h_{ij}^m + \left(1 - c_{ij}\right)\log\left(1 - h_{ij}^m\right). \tag{15}$$

The hash quantization loss is implemented using (12), which encourages the learned hash codes to be assigned values as close to $\{-1, 1\}$ as possible. We define the hash quantization loss as follows:

$$Q_{\text{hq}} = \sum_{m=1}^{M}\sum_{i=1}^{n}\sum_{j=1}^{K} \log\cosh\left(\left|h_{ij}^m\right| - 1\right). \tag{16}$$

Based on $\mathcal{L}$, C^{2}Hash can be optimized to learn hashing functions that directly encode multisource RS images into unified hash codes for efficient multisource RS image retrieval. The best hyperparameters are obtained through a grid search in this work. The detailed configuration of C^{2}Hash and experiment results will be discussed in Section IV.

E. Out-of-Sample Extension

C^{2}Hash consists of multiple hashing networks, each of which operates without sharing parameter weights. These networks can be independently used to extract hash codes for their corresponding image sources. To obtain the hash code for a query image from the mth image source, we can directly utilize the mth hashing network in the following manner:

$$h_q^m = \text{sign}\left(\tanh\left(g_m\left(f_m\left(x_q^m, W_z^m\right), W_h^m\right)\right)\right) \tag{17}$$

where x_q^m represents a query from the mth image source and h_q^m denotes the extracted hash code for x_q. These extracted hash codes can be utilized to retrieve similar images from different image sources.

IV. Experiments

A. Dataset

In this article, we propose a deep cross-modal hashing technique for multisource RS image retrieval, specifically focusing on scenarios with more than two image sources. However, to the best of our knowledge, there is currently no publicly available multisource RS image dataset for RS image retrieval. Existing datasets only comprise dual-source RS images. To address this limitation, we have constructed a novel MSRSI dataset. This dataset includes 15 distinct land-cover categories and encompasses five different types of RS image sources. The land-cover categories include industrial area, dense residential, sea and lake, river, bridge, rectangular farmland, circular farmland, forest, grassland, beach, wastewater plant, solar panel, desert, airport, and overpass. The sources are VHR, SAR, MSI, MAP, and PAN images.

To construct the MSRSI dataset, we collect these multisource RS images from various platforms, such as Google Maps. First, we downloaded VHR images from Google Maps and ArcGIS Maps [56] and recorded their geographical locations and acquisition dates. Our primary focus is on VHR images captured within the past five years. Each VHR image has dimensions of 640 × 640 pixels and consists of three spectral bands: red, green, and blue, with a spatial resolution of 1 m. Second, we obtained MSI, SAR, and PAN images from Google Earth Engine [57], a cloud computing-based RS platform. These images were selected based on the geographical locations of VHR images and the closest timestamps to VHR images. The MSI images were collected from Sentinel-2 satellite, covering 13 different spectral bands: B1, B2, B3, B4, B5, B6, B7, B8, B8A, B9, B10, B11, and B12. Each MSI image has dimensions of 64 × 64 pixels and contains less than 1% cloud cover. The SAR images were acquired from the Sentinel-1 satellite and included dual-polarized information channels (VV and VH) in interferometric wide swath (IW) mode. Each SAR image has dimensions of 64 × 64 pixels with a spatial resolution of 10 m. PAN images were downloaded from the Landsat 8 satellite and consisted of a single PAN band (B8). These PAN images measure 42 × 42 pixels and have a spatial resolution of 15 m. Finally, we collected MAP images from OpenStreetMap, a free and open geographic database that provides maps of the entire world. Each MAP image is an RGB image with dimensions of 640 × 640 pixels. To ensure

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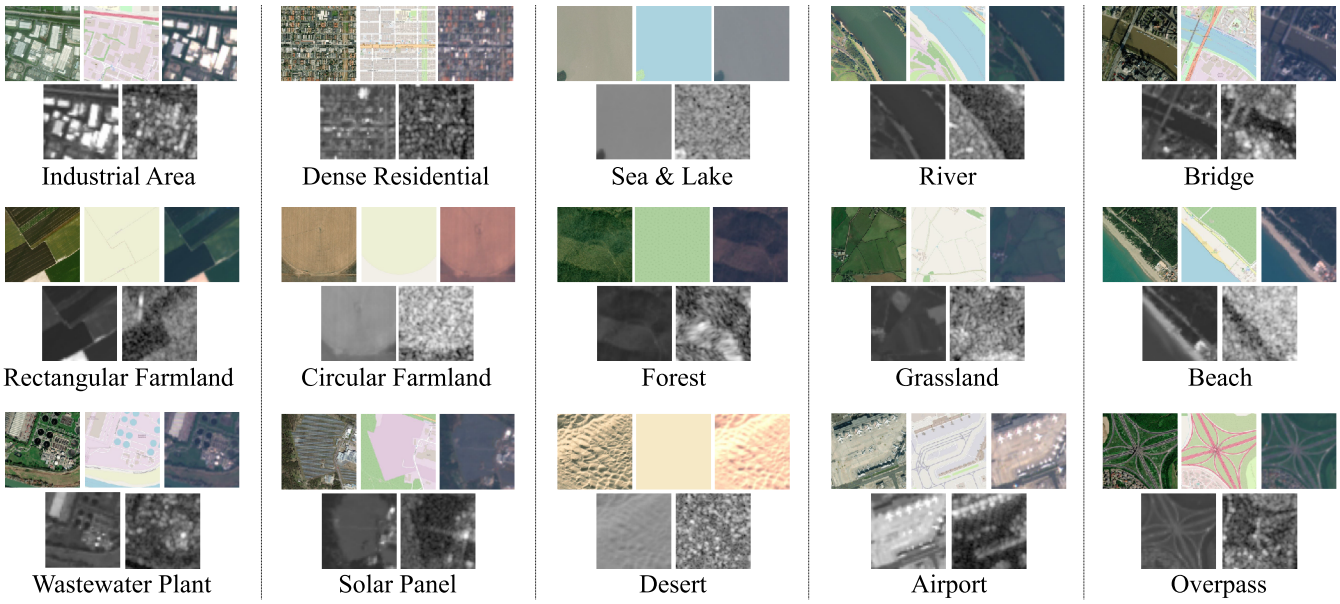

Fig. 3. Few examples from the MSRSI dataset. These examples show VHR, MAP, MSI, PAN, and SAR images from 15 land-cover categories. Each multisource RS image example shares the same land-cover category and was captured at the same geographic location.

the consistency of the dataset, we manually reviewed these multisource images and removed any images with inconsistent locations or categories.

After completing the aforementioned data processing procedures, a substantial number of multisource images were obtained. Subsequently, we manually annotated these images to create the multisource RS dataset. The final dataset consists of 150 000 images collected using five distinct image sources, categorized into 15 land-cover classes. Each image source contains 30 000 images, while each land-cover category comprises 2000 images. Fig. 3 shows a few examples of multisource RS images from the MSRSI dataset. Each image set displays the samples of the same land-cover category from five image sources captured at the same geographic location. To train and evaluate our model, we divide the dataset into test, validation, and training sets. The test set and validation set both contain 15 000 images, with 200 images per category and image source. The remaining 120 000 images make up the training set. During the evaluation process, the test or validation images are treated as query images, while the training images serve as database images. It is worth mentioning that the MSRSI dataset is publicly available.[1]

B. Experimental Setup

1) Implementation Details: Our C^{2}Hash method is implemented using the PyTorch framework. All our experiments are carried out on a workstation equipped with an Intel[2] Core[3] i9-10920X CPU running at 3.50 GHz, along with 2-GB GeForce RTX 3090 24-GB TURBO GPUs. C^{2}Hash employs stochastic gradient descent as our optimization algorithm, with a mini-batch size of 256. The initial learning rate is set to $10^{-1.5}$ and linearly decreases to 10^{-6} as the number of epochs increases. We conduct a total of 100 epochs in our experiment. C^{2}Hash mainly relies on two hyperparameters, namely, λ_0 and λ_1, which are set to $\lambda_0 = 5.0$ and $\lambda_1 = 1.0$, respectively.

2) Baselines: We compare C^{2}Hash with ten classical or state-of-the-art supervised cross-modal hashing methods, including four traditional shallow methods (IMH [39], LSSH [40], CMFH [41], and SCM [42]) and six recently proposed deep methods (DCMHN [38], DCMH [43], SIDHCNN [37], CMHH [44], DSMHN [45], and MsEspH [46]). Following the approach in [37] and [46], we use deep features extracted by a CNN model as input for the traditional shallow hashing methods. For the deep learning-based methods, raw image pixels are directly fed into the deep networks. To ensure a fair and consistent comparison, we employ the same network architecture to all deep hashing methods. In addition, the images from all sources are cropped to a size of 224 × 224 and normalized using standard deviation and mean values. Some image sources contain more than three bands. For example, MSI images consist of 13 spectral bands. We used convolutional neural networks to convert these images into three bands.

3) Evaluation Protocol: Four evaluation metrics are commonly used to assess the performance of image retrieval methods [9], [15], [37], [38], [46], [58]: MAP, precision–recall curves, top-N-precision ($P@N$) curves, and precision within Hamming radius R ($P@H{\leq}$R) curves. The definitions of these metrics are given as follows.

1) MAP represents the AP across a given set of query images. It is calculated as the mean of the AP scores, and the formula is given as follows:

$$\mathrm{MAP} = \frac{1}{\mathrm{Q}} \sum_{\mathrm{q}=1}^{\mathrm{Q}} \mathrm{AP}(\mathrm{x_q}) \tag{18}$$

where Q denotes the total number of queries and AP(x_q) represents the AP score for the query image x_q. The AP

[1] https://github.com/sunyuxi/C2Hash
[2] Registered trademark.
[3] Trademarked.

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TABLE I
COMPARISON OF MAP IN HAMMING RANKING FOR VARIOUS METHODS IN MULTISOURCE RS IMAGE RETRIEVAL

Task	Shallow				Deep						
	IMH	LSSH	CMFH	SCM	DCMHN	DCMH	SIDHCNN	CMHH	DSMHN	MsEspH	C^{2}Hash
MSI→VHR	0.1534	0.1951	0.1714	0.4925	0.7244	0.8529	0.9050	0.8865	0.8722	0.8852	**0.9191**
MSI→MAP	0.1593	0.1710	0.1653	0.4860	0.7697	0.8746	0.8949	0.8724	0.8615	0.8721	**0.9017**
MSI→PAN	0.0863	0.1179	0.0971	0.3773	0.6636	0.7076	0.8382	0.8738	0.8978	0.9026	**0.9208**
MSI→SAR	0.0244	0.0475	0.0576	0.2063	0.5704	0.7115	0.8365	0.8582	0.8715	0.8868	**0.9213**
VHR→MAP	0.1796	0.2136	0.1958	0.6172	0.7314	0.8290	0.9188	0.8887	0.8752	0.8805	**0.9295**
VHR→MSI	0.1475	0.1839	0.1673	0.5187	0.7584	0.8565	0.9471	0.9276	0.9146	0.9216	**0.9501**
VHR→PAN	0.1377	0.1701	0.1407	0.4572	0.6207	0.7307	0.8625	0.8881	0.9057	0.9140	**0.9497**
VHR→SAR	0.0303	0.0617	0.0754	0.2415	0.5291	0.7234	0.8580	0.8710	0.8748	0.8958	**0.9504**
PAN→VHR	0.1469	0.1792	0.1444	0.4381	0.5518	0.6360	0.7185	0.7210	0.7344	0.7369	**0.7772**
PAN→MAP	0.1361	0.1578	0.1510	0.4279	0.5853	0.6435	0.7121	0.7092	0.7308	0.7271	**0.7625**
PAN→MSI	0.0874	0.1202	0.0976	0.3769	0.6098	0.6557	0.7301	0.7443	0.7671	0.7620	**0.7785**
PAN→SAR	0.0153	0.0459	0.0513	0.1889	0.4685	0.5951	0.7172	0.7020	0.7260	0.7364	**0.7786**
SAR→VHR	0.0333	0.0734	0.0759	0.2351	0.4818	0.5952	0.6704	0.6860	0.6751	0.6952	**0.7452**
SAR→MAP	0.0294	0.0686	0.0834	0.2293	0.5207	0.6098	0.6718	0.6775	0.6753	0.6863	**0.7323**
SAR→MSI	0.0262	0.0526	0.0596	0.2024	0.5412	0.6221	0.6806	0.7104	0.7169	0.7238	**0.7470**
SAR→PAN	0.0164	0.0554	0.0472	0.1858	0.4788	0.5651	0.6693	0.6817	0.6962	0.7115	**0.7468**
MAP→VHR	0.1812	0.2087	0.1902	0.6284	0.7598	0.8796	0.9409	0.9133	0.8998	0.9087	**0.9537**
MAP→MSI	0.1572	0.1558	0.1556	0.5148	0.8383	0.9421	0.9550	0.9391	0.9385	0.9367	**0.9560**
MAP→PAN	0.1251	0.1445	0.1342	0.4485	0.6932	0.7438	0.8657	0.9055	0.9194	0.9272	**0.9555**
MAP→SAR	0.0263	0.0566	0.0727	0.2307	0.5938	0.7388	0.8641	0.8852	0.8842	0.9101	**0.9560**
Average	0.0950	0.1240	0.1167	0.3752	0.6245	0.7257	0.8128	0.8171	0.8218	0.8310	**0.8666**

is defined using the following formula:

$$\mathrm{AP}(x_q) = \frac{1}{R}\sum_{r=1}^{R} P\left(x_q, r\right)\theta(r) \tag{19}$$

where R is the number of ground-truth images in the database for each query and $P(x_q, r)$ denotes the precision of the rth retrieved image. The retrieval process involves recalling and sorting the images in ascending order based on the Hamming distance calculated between the query image x_q and the database images. In this work, we set $R = 1600$ to calculate the MAP of all methods. In additionally, the indicator function $\theta(r)$ is utilized to determine whether the rth retrieved image shares at least one common label with the query x_q. If it does, $\theta(r)$ equals 1; otherwise, $\theta(r)$ is set to 0.

2) Precision–recall curves illustrate how precision varies across different recall values. These precision and recall values are computed using various searching Hamming radii. A greater area under the precision–recall curve indicates superior retrieval performance.
3) Top-N-precision ($P@N$) is a metric that measures the precision of the top N retrieved images. Given a set of retrieved images that include images similar to the query image, $P@N$ represents the proportion of similar images to the query image among the top N retrieved images. It can be calculated using the following formula:

$$P@N = \frac{k}{N} \tag{20}$$

where N represents the total number of retrieved images and k is the number of images similar to the query image within the retrieved image set. $P@N$ indicates how many images among the top N retrieved images are similar to the query image.
4) Precision within a Hamming radius ($P@H \leq R$) measures the precision of retrieved images whose Hamming distance from the query image falls within a specified radius, R. Hamming distance quantifies the number of different bit positions between two hash codes. The choice of R can vary based on specific application requirements or retrieval system needs. Common values for R include 0, 1, 2, or 3. Evaluating $P@H \leq R$ is crucial for assessing the effectiveness of retrieval algorithms that employ binary hash codes, as it provides valuable insights into precision under different distance constraints. Utilizing binary hash codes offers the advantage of constant time complexity ($O(1)$) for the Hamming ranking process, ensuring efficiency in large-scale retrieval tasks.

C. Experimental Results

1) Comparison With Baseline Methods: We present a comprehensive quantitative comparison of our C^{2}Hash with existing cross-modal hashing methods on the MSRSI dataset and report the results in Table I. Table I shows the MAP scores of all methods using 32-bit hash codes for image retrieval across different image sources. In Table I, "Shallow" refers to traditional shallow hashing methods that utilize deep features, while "Deep" indicates deep hashing methods based on neural networks. Based on the results presented in Table I, we can make the following observations.

First, shallow hashing methods demonstrate inferior performance compared to deep learning methods, even when utilizing deep features as input. For example, DCMHN

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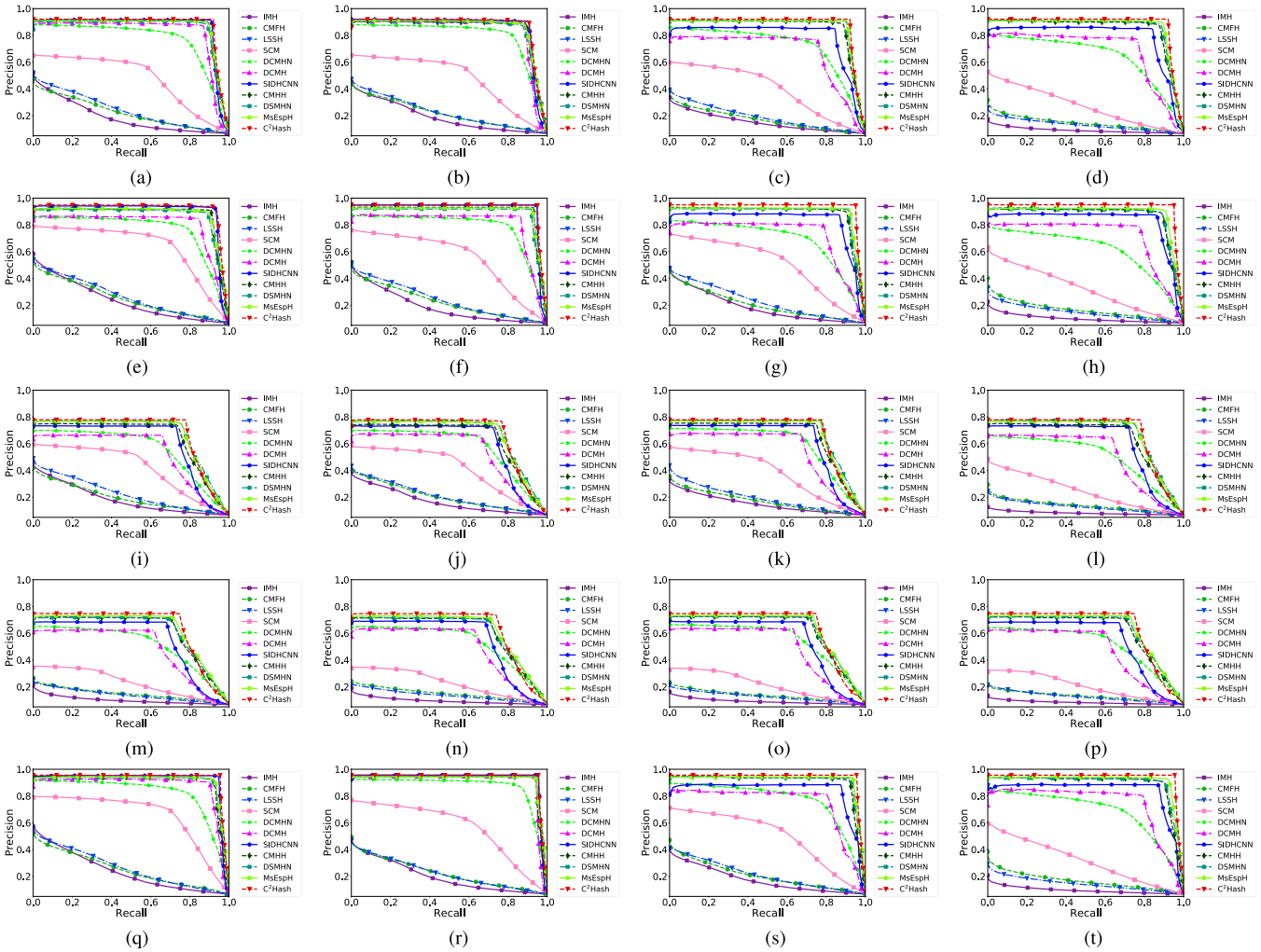

Fig. 4. Precision–recall curves on the MSRSI dataset. (a) MSI → VHR. (b) MSI → MAP. (c) MSI → PAN. (d) MSI → SAR. (e) VHR → MAP. (f) VHR → MSI. (g) VHR → PAN. (h) VHR → SAR. (i) PAN → VHR. (j) PAN → MAP. (k) PAN → MSI. (l) PAN → SAR. (m) SAR → VHR. (n) SAR → MAP. (o) SAR → MSI. (p) SAR → PAN. (q) MAP → VHR. (r) MAP → MSI. (s) MAP → PAN. (t) MAP → SAR.

presents an absolute improvement of 24.93% in the average MAP of all image retrieval tasks compared to SCM. This is because shallow methods separately learn feature representations and hashing functions for RS images, which restricts the discrimination capability of the generated hash codes. In contrast, deep hashing methods incorporate convolutional neural networks as feature representation layers and include a hashing layer to jointly perform representation learning and hash coding. Consequently, all deep methods consistently outperform shallow methods.

Second, our C^2Hash demonstrates superior retrieval performance across various cross-modal image retrieval tasks compared to other methods. For example, compared to MsEspH, C^2Hash achieves significant improvements of 5.47%, 4.87%, 2.17%, and 5.73% in PAN→VHR, PAN→MAP, PAN→MSI, and PAN→SAR tasks, respectively. This is because our C^2Hash directly preserves semantic similarity and minimizes the modal discrepancies among all multisource RS images. In contrast, existing methods primarily focus on reducing discrepancies between two modalities using pairwise or triplet data, which may not yield optimal hash codes for multisource RS images. Overall, C^2Hash outperforms all baseline methods in image retrieval tasks involving multiple image sources.

In addition, we conduct further analysis regarding the impact of image source selection on retrieval results. As shown in Table I, our proposed method performs well when using VHR, MSI, and MAP images as query images. On the contrary, the retrieval performance of our method is less satisfactory when SAR and PAN images are used as query images. The reason behind this discrepancy lies in the inherent characteristics of RS images. Specifically, SAR images tend to have fewer texture features, while PAN images have lower spatial resolution compared to VHR, MSI, and MAP images. Consequently, SAR and PAN images pose greater challenges for our proposed method in terms of image retrieval. For example, the average mAP between SAR and PAN image sources is found to be 0.7627. In contrast, the average mAP between VHR and MSI images is significantly higher at 0.9346, reflecting a considerable absolute improvement of 17.19%. These results clearly emphasize the crucial role of texture features and spatial resolution in determining the quality of retrieval results. In conclusion, our experimental results

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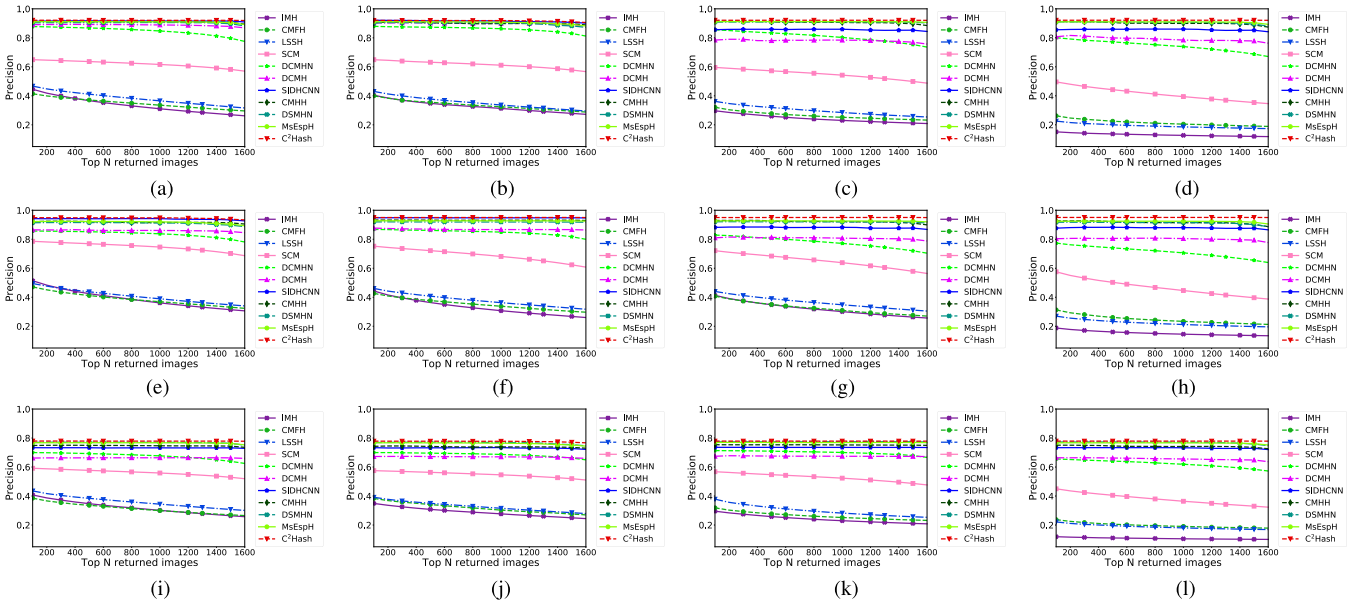

Fig. 5. Top-N precision curves on the MSRSI dataset for all methods, treating MSI, VHR, and PAN images as query images. (a) MSI → VHR. (b) MSI → MAP. (c) MSI → PAN. (d) MSI → SAR. (e) VHR → MAP. (f) VHR → MSI. (g) VHR → PAN. (h) VHR → SAR. (i) PAN → VHR. (j) PAN → MAP. (k) PAN → MSI. (l) PAN → SAR.

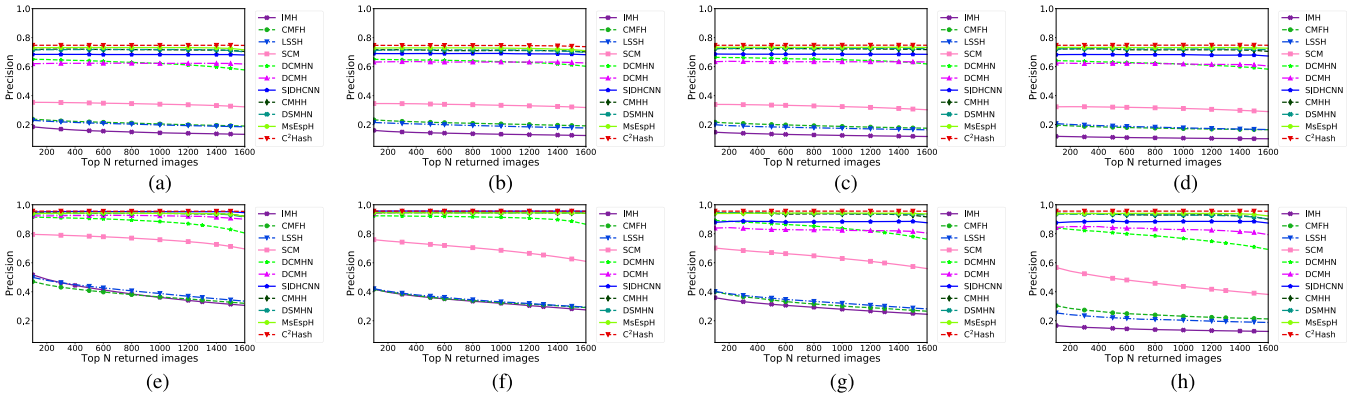

Fig. 6. Top-N precision curves on the MSRSI dataset for all methods, treating SAR, and MAP images as query images. (a) SAR → VHR. (b) SAR → MAP. (c) SAR → MSI. (d) SAR → PAN. (e) MAP → VHR. (f) MAP → MSI. (g) MAP → PAN. (h) MAP → SAR.

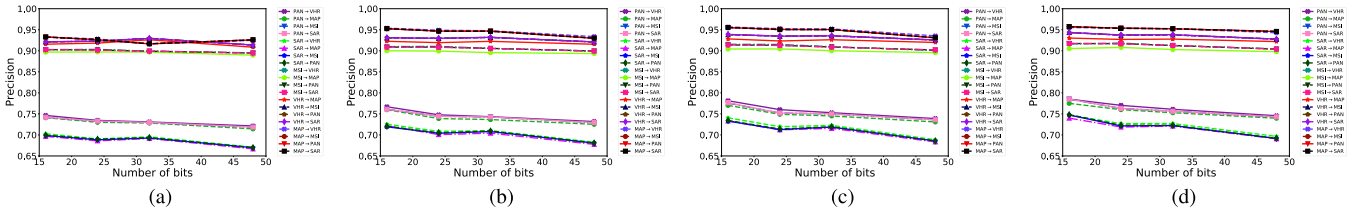

Fig. 7. Precision curves within Hamming radius R ($P@H \leq R$), $R \in \{0,1,2,3\}$. (a) $P@H \leq 0$. (b) $P@H \leq 1$. (c) $P@H \leq 2$. (d) $P@H \leq 3$.

show that VHR, MSI, and MAP image sources are more favorable choices to achieve superior retrieval performance within our proposed dataset.

Fig. 4 presents the precision–recall curves for all methods across various retrieval tasks. We can see that our C^{2}Hash consistently outperforms the other methods in these tasks, as evidenced by its larger area under the curve. For example, in Fig. 4(a) for the MSI→VHR task, the precision–recall curve of C^{2}Hash exhibits the largest area, indicating its superior performance and robustness compared to the other methods. In addition, deep hashing methods consistently achieve higher precision than shallow hashing methods at lower recall levels across different tasks. For example, in Fig. 4(c) for the MSI→PAN task, there is a significant precision margin between C^{2}Hash and LSSH when the recall value is 0.2. In summary, C^{2}Hash outperforms all baseline methods across different tasks.

The top-N-precision curves for all methods in different tasks can be observed in Figs. 5 and 6. Fig. 5 shows the results when query images are sourced from MSI, VHR, and

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TABLE II
COMPARISON OF MAP BETWEEN MSESPH AND C^{2}HASH AT DIFFERENT HASH BIT LENGTHS

Task	MsEspH				C^{2}Hash			
	16 bits	24 bits	32 bits	48 bits	16 bits	24 bits	32 bits	48 bits
MSI→VHR	0.8788	0.8764	0.8852	0.8908	0.9156	0.9214	0.9191	0.9156
MSI→MAP	0.8713	0.8621	0.8721	0.8714	0.8966	0.9044	0.9017	0.8935
MSI→PAN	0.8990	0.8911	0.9026	0.9084	0.9180	0.9234	0.9208	0.9192
MSI→SAR	0.8656	0.8682	0.8868	0.8815	0.9182	0.9241	0.9213	0.9198
VHR→MAP	0.8812	0.8821	0.8805	0.8755	0.9212	0.9242	0.9295	0.9161
VHR→MSI	0.9185	0.9238	0.9216	0.9222	0.9443	0.9453	0.9501	0.9437
VHR→PAN	0.9099	0.9147	0.9140	0.9145	0.9436	0.9446	0.9497	0.9431
VHR→SAR	0.8738	0.8871	0.8958	0.8818	0.9443	0.9453	0.9504	0.9437
PAN→VHR	0.7277	0.7445	0.7369	0.7355	0.7849	0.7780	0.7772	0.7759
PAN→MAP	0.7224	0.7304	0.7271	0.7191	0.7673	0.7624	0.7625	0.7560
PAN→MSI	0.7564	0.7673	0.7620	0.7613	0.7880	0.7803	0.7785	0.7797
PAN→SAR	0.7144	0.7313	0.7364	0.7221	0.7879	0.7803	0.7786	0.7797
SAR→VHR	0.6806	0.6840	0.6952	0.6823	0.7451	0.7390	0.7452	0.7248
SAR→MAP	0.6804	0.6714	0.6863	0.6691	0.7302	0.7256	0.7323	0.7085
SAR→MSI	0.7081	0.7118	0.7238	0.7134	0.7474	0.7412	0.7470	0.7286
SAR→PAN	0.7004	0.6970	0.7115	0.7017	0.7471	0.7406	0.7468	0.7282
MAP→VHR	0.8978	0.9148	0.9087	0.9032	0.9544	0.9566	0.9537	0.9521
MAP→MSI	0.9269	0.9439	0.9367	0.9265	0.9573	0.9593	0.9560	0.9565
MAP→PAN	0.9220	0.9317	0.9272	0.9207	0.9569	0.9586	0.9555	0.9560
MAP→SAR	0.8851	0.9027	0.9101	0.8877	0.9573	0.9593	0.9560	0.9565
Average	0.8210	0.8268	0.8310	0.8244	**0.8663**	**0.8657**	**0.8666**	**0.8598**

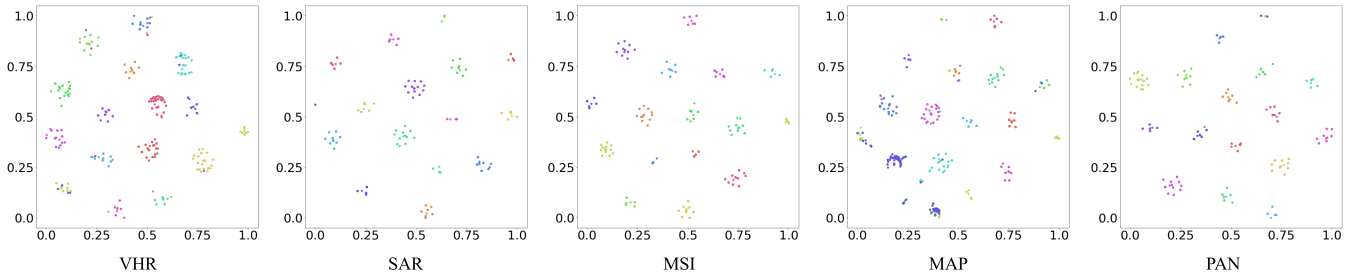

Fig. 8. t-SNE visualization of hash codes extracted by C^{2}Hash from VHR, SAR, MSI, MAP, and PAN images. The colors represent different categories based on ground-truth labels.

PAN, while Fig. 6 shows the results using SAR and MAP images as query images. From Figs. 5 and 6, we observe that C^{2}Hash consistently achieves higher precision than other methods across different N values for various image retrieval tasks, indicating its superior performance in retrieving top items. For example, in the MSI→PAN task shown in Fig. 5(c), C^{2}Hash exhibits the highest top-N-precision curve among all methods. Furthermore, C^{2}Hash consistently maintains high precision throughout the entire range of N values, demonstrating its consistent accuracy in retrieving top items. Overall, C^{2}Hash outperforms all comparison methods in retrieval performance.

We present the MAP results of C^{2}Hash and MsEspH for different hash bit lengths across various image retrieval tasks in Table II. We can see that C^{2}Hash consistently outperforms MsEspH across different hash bit lengths. For example, compared to MsEspH, C^{2}Hash demonstrates significant improvements in average MAP, with increments of 5.52%, 4.70%, 4.28%, and 4.29% for 16-, 24-, 32-, and 48-bit hash codes, respectively. The superior performance of C^{2}Hash can be attributed to its capability to effectively handle discrepancies among all image sources and generate unified hash codes for multisource images. Furthermore, C^{2}Hash exhibits superior retrieval performance compared to MsEspH in various image retrieval tasks. For example, with 16-bit hash codes, C^{2}Hash achieves relative improvements of 4.19%, 2.90%, 2.11%, and 6.08% in the MSI→VHR, MSI→MAP, MSI→PAN, and MSI→SAR tasks, respectively, when compared to MsEspH. Overall, C^{2}Hash consistently outperforms MsEspH in terms of retrieval performance across different hash bit lengths.

Fig. 7 shows the precision results of C^{2}Hash when considering the Hamming radius R, which varies within the set $\{0, 1, 2, 3\}$. We can see that C^{2}Hash consistently achieves satisfactory results with $P@H \leq R$ across different radii, even when using varying numbers of bits. For example, in Fig. 7(b) and (d), the MAP scores of the MAP→SAR remain stable regardless of the code length. As the hash codes become longer, the Hamming space becomes sparser, resulting in fewer data points falling within the Hamming ball of radius R. However, C^{2}Hash effectively utilizes consistency centers to group similar multisource images within the

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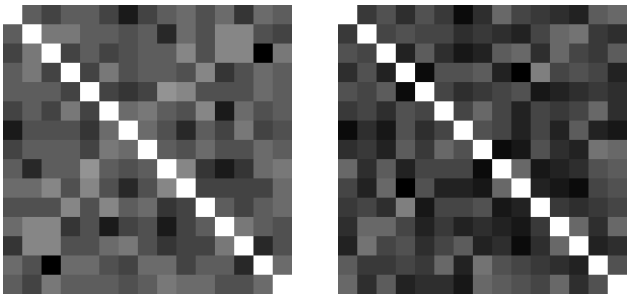
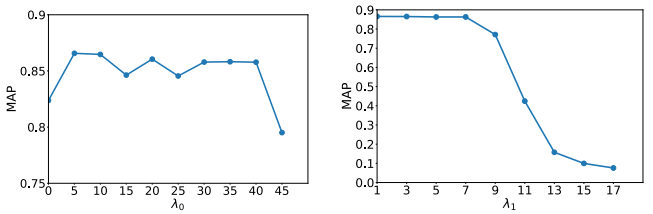


(a) (b)

Fig. 9. Visualization of the consistency centers generated by C^{2}Hash. Each row or column corresponds to a consistency center. The cells in the visualization display the inner product values between two consistency centers. Darker colors indicate lower inner product values, suggesting a larger Hamming distance between the consistency centers. (a) Visualization of consistency centers on 24-bit hash codes. (b) Visualization of consistency centers on 32-bit hash codes.

TABLE III
ABLATION TEST RESULTS IN TERMS OF MAP OBTAINED BY DISCARDING CONSISTENCY CENTERS (W/O CON-CENTERS)

Method	16 bits	24 bits	32 bits	48 bits
w/o con-centers	0.8276	0.8111	0.8122	0.8029
C^{2}Hash	0.8663	0.8657	0.8666	0.8598



(a) (b)

Fig. 10. Sensitivity analysis of (a) λ_0 and (b) λ_1 on the MSRSI dataset.

Hamming ball. Consequently, C^{2}Hash demonstrates consistent performance across different code lengths and Hamming radii.

2) Visualization of Hash Code Distance: We present the visualization of t-distributed stochastic neighbor embedding (t-SNE) results for C^{2}Hash using 32-bit hash codes in Fig. 8, representing different image sources. t-SNE is a statistical method that allows us to visualize high-dimensional data by mapping each data point to a 2-D map. In Fig. 8, different colors indicate distinct ground-truth class labels in each t-SNE plot. We observe that C^{2}Hash produces high-quality hash codes for each image source, exhibiting cohesion among similar images and separation between dissimilar images. This implies that C^{2}Hash is more likely to retrieve the images similar to query images when searching across multiple data sources. This may be because C^{2}Hash uses consistency centers to deal with the discrepancies among multiple image sources. In conclusion, C^{2}Hash can generate high-quality hash codes that effectively distinguish between similar and dissimilar RS images.

3) Qualitative Results: C^{2}Hash utilizes consistency centers to minimize discrepancies among various image sources, emphasizing the importance of generating high-quality centers for multisource RS images. The quality of these consistency centers is characterized by a Hamming distance between them. Orthogonal constraints, as denoted by (7), can assist in generating high-quality consistency centers. In our experiments, we employ a fully connected layer to learn all consistency centers. To evaluate the quality of the acquired centers, we visualize their inner product for both 24- and 32-bit hash codes, as shown in Fig. 9(a) and (b), respectively. Each pixel in Fig. 9(a) and (b) is the result of calculating the inner product of two consistency centers, which is then scaled to a range of $[0, 255]$ for visualization. The darker colors in Fig. 9(a) and (b) correspond to smaller inner products. Fig. 9(a) and (b) shows that most inner products between consistency centers are close to 0. A smaller inner product implies a larger Hamming distance between two centers, indicating that similar multisource images are positioned closely, while dissimilar ones are situated further apart. As a result, the hash codes generated possess a high level of discrimination within the Hamming space. In summary, C^{2}Hash can generate high-quality consistency centers, thereby facilitating the generation of effective hash codes for multisource RS images.

D. Ablation Study

C^{2}Hash incorporates consistency centers into its objective function design. To investigate the impact of different design choices in our proposed approach, we conduct an ablation study where we exclude the consistency centers (referred to as "w/o con-centers"). In the "w/o con-centers" method, we omit consistency centers and focus solely on optimizing C^{2}Hash using class labels. The ablation test results, presented in Table III, summarize the average MAP for image retrieval from various image sources. We observe that consistency centers significantly enhance the performance of C^{2}Hash across different hash codes. For example, compared to the "w/o con-centers" method, C^{2}Hash achieves notable improvements of 4.67%, 6.73%, 6.70%, and 7.09% for 16-, 24-, 32-, and 48-bit hash codes, respectively. In summary, consistency centers are crucial for significantly improving the retrieval performance of C^{2}Hash.

E. Parameter Analysis

We investigate the impact of hyperparameters λ_0 and λ_1 on C^{2}Hash. The role of λ_0 is to control the multisource hashing consistency loss, while λ_1 controls the hash quantization loss. In our experiments, we empirically set λ_0 to 5 and λ_1 to 1. To analyze the effect of λ_0, we keep λ_1 fixed and vary λ_0 to measure the average MAP of C^{2}Hash across various tasks using 24-bit hash codes. The results, shown in Fig. 10(a), show that the MAP score initially increases as the values of λ_0 increase and then gradually decreases. When λ_0 becomes excessively large, C^{2}Hash overly emphasizes the differences among multiple image sources and pays less attention to semantic similarity. Consequently, C^{2}Hash cannot generate effective hash codes for multisource RS images. Similarly, we evaluate the impact of λ_1 on C^{2}Hash, and the results are

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presented in Fig. 10(b). It is observed that the MAP scores of C^{2}Hash slightly decrease as λ_1 increases, with a significant decline occurring when λ_1 exceeds 9. When assigning a high weight to the hash quantization loss, C^{2}Hash primarily focuses on hash bit distribution. As a result, C^{2}Hash cannot effectively learn semantic similarity and reduce the differences among multisource RS images.

F. Downstream Applications

The proposed multisource RS image retrieval technique offers a wide range of potential applications [1]. One notable application is its significance in disaster rescue and assessment scenarios. For example, in maritime search and rescue surveillance [59], C^{2}Hash can be employed to identify similar ships captured in SAR images taken at night, facilitating the search across different time frames and satellite sources. Similarly, during emergency disaster assessment [60], C^{2}Hash can be used to locate and assess regions with greater damage by utilizing diverse satellite images.

V. Conclusion

In this article, we introduce C^{2}Hash, a novel deep cross-modal hashing approach (based on consistency centers) for multisource RS image retrieval. Our newly proposed method comprises three main components: a deep multibranch hashing network, a label hashing network, and an objective function based on consistency centers. The deep multibranch network consists of multiple independent hashing networks, each designed to process an individual image source. By leveraging this network, we can effectively encode multisource RS images into unified hash codes. The label hashing network is applied to generate consistency centers. Our objective function utilizes these centers to optimize C^{2}Hash, incorporating three loss components: a center-based similarity preserving loss, a multisource hashing consistency loss, and a hash quantization loss. The similarity preserving loss learns semantic information across different image sources by utilizing class labels and the generated centers. The hashing consistency loss utilizes the consistency centers to minimize discrepancies among multisource RS images. The hash quantization loss is used to maintain the distribution of hash codes. Furthermore, we contribute a publicly available multisource RS image dataset named MSRSI. We conduct extensive experiments on the dataset to evaluate the effectiveness of C^{2}Hash. Compared with various state-of-the-art methods, C^{2}Hash consistently achieves the best results in terms of image retrieval between different image sources.

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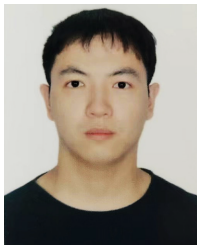

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